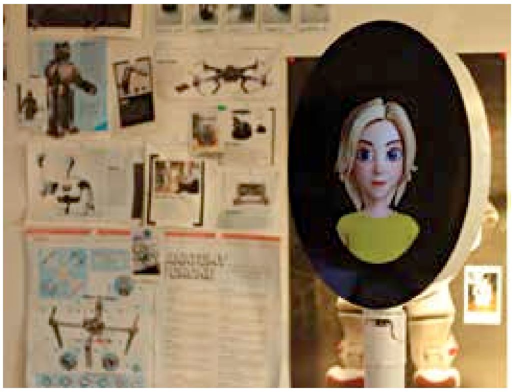
Cambridge presented a paper that shook the robotic community. An unassuming robotic arm could build smaller robots, select the ones working properly and decommission the rest. These smaller robots could, in turn, build themselves into higher grade robots, and continue the process. This emulation led to a direct comparison with mankind – mothers giving birth to babies, choosing the ones fittest for survival, thereby continuing the process of evolution. The team has presented its demonstration of morphological evolution with model-free phenotype development, terming the instructions fed to the robotic arm as 'genes', which would pass the same into the 'baby robots', much like natural evolution.

While this is a far cry from complete automation and evolution, it's a stepping stone for the future. With the advent of deep neural learning, robots are being



Self aware robots have formed the basis for Sci-Fi

imparted with neural action reflex, computer vision, speech and face recognition. The principle is a breakaway from recognition-based learning. Machines are being taught a representational method of recognition and are even able to see, hear and speak by themselves.

BLARING SIGNALS

Elon Musk has been one of the earliest to acknowledge the havoc that machine learning and artificial intelligence can wreak. Deep neural learning is the answer to the questions on how can robots echo the divide-and-rule model of humans, how they can establish similar personal interactions and relationships.

Machine learning and artificial intelligence technologies are combining to steer forward not only the future of machines, but they are already shaping the way mankind is evolving further, after having reached a considerable stage of intelligence and advancement. First signs of warning? While the blaring signals have not yet been set off, progress for progress's sake must be curbed, to find the perfect time, place and moment for everything.

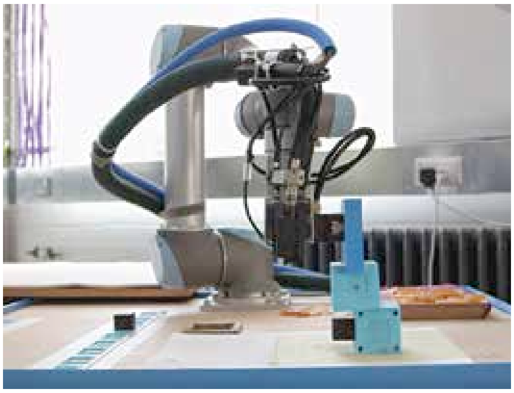
For now, there is no slowing down. With the scope of computing set to enter nano-technology and the realm of quantum computing, machine learning has the highest, fastest window of growth in future. While artificial intelligence focuses on teaching robots how to recognise situations and make decisions based on the same, machine learning is aiding the process by which robots can form a better understanding of the social model that we have carved for ourselves.

THE SCARY BIT

Teaching our own creations a few of our own tricks may be a big part of our future, and actually taking forward our research and forays into numerous other fields – astronomy, for instance. But, the question of AI ethics come to the forefront – robots can be programmed much (un)like ASIMO to respond to humans.

With advancement in realms of social intelligence, the creation of individual robot societies are highly probable. Matching algorithms may bring forth similar robots, who can in turn emulate our services to set up a deeply aware and connected robot collective. But, what happens when the clan of evil robots decide to attack? A recent factory incident, where a man was crushed to death by a robot, sparked fears of a probable future. Although this incident was an accident, how long will we have to wait, until the machines add violence and destruction to their deep neural networks?

The evolution of mankind is now deeply ingrained into automation and smart machines. Self-driving cars and retina-recognising smartphones are fast becoming reality, and along with them so are machines that can fully emulate our walk, speech, sight and partially thought as well. Back in 1950s, in one of the first trysts with machine learning and artificial intelligence, Alan Turing played the 'Imitation Game', suggesting a computer to study human inputs and guess the gender of the person using it. After a number of decades, research laboratory prototypes are increasingly evolving, bringing the doomsday project closer.



They are out to get us. Skynet has gone sentient

Machines are learning and gathering more data about environmental interactions, business tasks, decision-making and other cognitive features. One of the world's first personal robot using machine learning and deep networks, named the Personal Robot, can be a complete companion for homes, featuring natural language understanding, smart connectivity, and the abilities of a multipurpose assistant. With such services, we are spending increasing amounts of time with entities wired to respond in a particular way, forgetting much about the human touch of care that machine learning technology is yet to cover. And yet, it is we who are imparting lessons of social coexistence to the robots, leaving a lot for the future to decide.

ONCE UPON A FUTURE...

Machine learning in today's day and age is more closer to a form of iterative self-versioning than true evolution. To understand what evolution means, let us extrapolate what effects it could have on mankind. Perhaps millennia from now our genes would be encoded to stop at the sight of a red light because as a species we've been passing down that learning to our progeny for generations. In the context of machines, only when they are able to self replicate and pass on their learnings to their progeny will they truly evolve. Even if that learning is not based on understanding – that's where the AI bit kicks in – it is still evolution in the truest sense of the word.